

Introduction

A traditional pitching line tells us what happened, but it does not always explain why. A pitcher can post a strong result because he located well, faced a favorable group of hitters, or benefited from a short sample. The opposite can happen just as easily. My goal with this Stuff+ model was to isolate the quality of the pitch itself as much as the available Cape Cod Baseball League tracking data would allow. The final score is meant to answer a practical question: if this pitch were thrown through a neutral set of locations and handedness matchups, how much swing and miss would its velocity, movement, release traits, and angles be expected to create?

My model is not intended to replace scouting or tell the entire story of a pitcher. Command, sequencing, count context, deception, durability, and contact management still matter. Instead, Stuff+ gives baseball analytics and player development staffs a common scale for discussing raw pitch characteristics. It can point analysts toward an unusual pitch, help coaches track whether a shape is holding through an outing, and provide another piece of evidence when evaluating an arsenal.

Data and Target Definition

The model uses pitch level TrackMan data from the 2025 and 2026 CCBL seasons. Before modeling, duplicate spreadsheet columns are combined, PitchUID values are used to remove duplicated pitches, pitch names are standardized, and pitcher and batter handedness are cleaned. Four seam fastballs, sinkers, two seam fastballs, cutters, sliders, curveballs, changeups, and splitters are kept as separate pitch groups.

The outcome is whiff probability on swings. Swinging strikes and blocked swinging strikes count as whiffs; fouls and balls put in play count as contact. Called balls and called strikes are not treated as failed whiff opportunities. The model is trained on the pitches where a hitter actually made a swing decision, then the fitted model is used to score every eligible pitch. As a result, the target is more closely tied to bat missing ability while the final reports can still cover a pitcher's complete tracked arsenal.

MyPitchType is the model's pitch classification because it reflects a cleaned pitch tag I want the pitch evaluated against. TaggedPitchType is preserved separately for the final reports. That means a pitch is scored using its intended model family but grouped under the tag that coaches and staff expect to see on the report. This separation avoids changing the model definition just to make the display look familiar.

Model Inputs

The modeling process begins with two related XGBoost classifiers. The context only model uses plate height, plate side, pitch type, pitcher handedness, and batter side. It establishes how much of the whiff outcome can be explained by where a pitch was thrown and the basic matchup. The full model adds the measured traits that describe the pitch like release and zone velocity, spin rate, spin axis, induced vertical break, horizontal break, total movement magnitude, release height, release side, extension, effective velocity, velocity loss, and location adjusted release and approach angles.

Spin axis is encoded with sine and cosine terms so that the model handles it as a circular measurement. A direction near 359 degrees should be treated as close to one degree, not as the opposite end of a linear scale. Just as important, the model does not manufacture active spin, gyro, or spin efficiency labels from movement and axis values that cannot physically support those conclusions. It uses the observed measurements directly and leaves more specific aerodynamic claims out of the score.

Missing numeric values are median imputed inside the modeling pipeline, with missingness indicators added automatically. Categorical features are imputed and one hot encoded with unknown levels allowed. The XGBoost models use shallow trees, a low learning rate, row and column subsampling, and both L1 and L2 regularization. Those settings are meant to reduce the chance that the model memorizes small CCBL samples or builds its score around one unusual pitcher.

Removing Location from the Final Score

A pitch quality model can accidentally become a location model even when plate coordinates are not the main focus. Approach angles and release angles change with the path needed to reach a particular part of the zone. If those raw angles are passed directly into a model, some of the location effect can remain hidden inside supposedly intrinsic features. To limit that problem, the model fits a separate quadratic ridge adjustment for each pitch type and pitcher handedness. Vertical angles are adjusted relative to plate height, while horizontal angles are adjusted relative to plate side. The adjusted value keeps the portion of the angle that is unusual after accounting for its expected location.

Location is still included during model training because it helps the model learn the difference between a pitch's physical traits and the context in which the swing occurred. The location neutral score is produced afterward through counterfactual prediction. Each pitch is copied across 18 neutral situations which are three plate heights, three horizontal locations, and both batter sides. The model predicts whiff probability in every situation, and those predictions are averaged. This approach asks the same pitch to travel through a

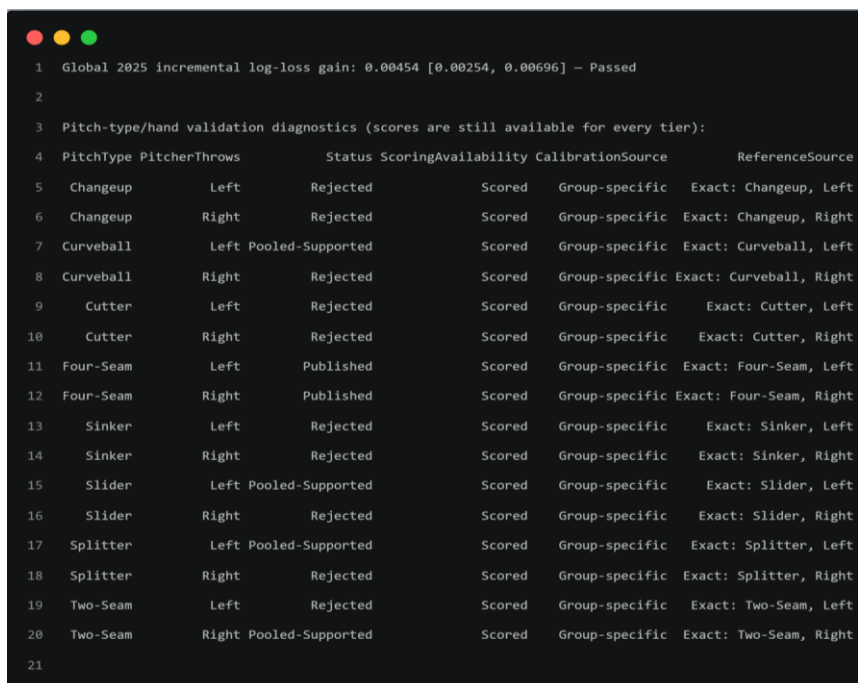
balanced set of contexts rather than rewarding the exact location where it happened to be thrown.

Validation and Leakage Control

The largest technical priority in this rebuild was making sure the validation process matched the way the model would be used. The 2025 season is the training season and 2026 is left untouched as a true seasonal holdout. Within 2025, GroupKFold validation holds out entire pitchers. A pitch from a held out pitcher cannot appear in the training portion of that fold. This is stricter than a random pitch split and gives a better estimate of whether the model can evaluate pitchers it has never seen.

Every learned preprocessing step is fit inside the fold. The angle adjustments are estimated only from training pitchers and then applied to the held out pitchers. Imputation, one hot encoding, and the XGBoost model are also fit only on the training side. The final 2026 predictions use models and angle adjusters fit on 2025 data. This structure removes the pitcher level and league wide leakage that can occur when transformations are calculated on the complete dataset before cross-validation.

The context only and full models are evaluated together. The main gate is incremental log loss gain. The full model should predict swing outcomes more accurately than location and matchup context alone. The code bootstraps pitchers, rather than individual pitches, to create uncertainty intervals around that gain. It also tracks AUC, precision recall AUC, log loss, Brier score, and expected calibration error. A pitch type and hand group can be marked Published when it clears the strict sample and confidence gates, Pooled Supported when the estimated signal is positive but the group interval is inconclusive, or Rejected when the evidence is not sufficient. The tier is a confidence label, not a reason to hide the score; eligible pitches remain available in the reports.



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1 Global 2025 incremental log-loss gain: 0.00454 [0.00254, 0.00696] - Passed
2
3 Pitch-type/hand validation diagnostics (scores are still available for every tier):
4 PitchType PitcherThrows      Status ScoringAvailability CalibrationSource      ReferenceSource
5 Changeup      Left      Rejected      Scored      Group-specific      Exact: Changeup, Left
6 Changeup      Right     Rejected      Scored      Group-specific      Exact: Changeup, Right
7 Curveball     Left     Pooled-Supported      Scored      Group-specific      Exact: Curveball, Left
8 Curveball     Right    Rejected      Scored      Group-specific      Exact: Curveball, Right
9 Cutter        Left     Rejected      Scored      Group-specific      Exact: Cutter, Left
10 Cutter        Right    Rejected      Scored      Group-specific      Exact: Cutter, Right
11 Four-Seam     Left     Published      Scored      Group-specific      Exact: Four-Seam, Left
12 Four-Seam     Right    Published      Scored      Group-specific      Exact: Four-Seam, Right
13 Sinker        Left     Rejected      Scored      Group-specific      Exact: Sinker, Left
14 Sinker        Right    Rejected      Scored      Group-specific      Exact: Sinker, Right
15 Slider        Left     Pooled-Supported      Scored      Group-specific      Exact: Slider, Left
16 Slider        Right    Rejected      Scored      Group-specific      Exact: Slider, Right
17 Splitter      Left     Pooled-Supported      Scored      Group-specific      Exact: Splitter, Left
18 Splitter      Right    Rejected      Scored      Group-specific      Exact: Splitter, Right
19 Two-Seam      Left     Rejected      Scored      Group-specific      Exact: Two-Seam, Left
20 Two-Seam      Right    Pooled-Supported      Scored      Group-specific      Exact: Two-Seam, Right
21
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PitchType	PitcherThrows	Status	ScoringAvailability	CalibrationSource	ReferenceSource
Changeup	Left	Rejected	Scored	Group-specific	Exact: Changeup, Left
Changeup	Right	Rejected	Scored	Group-specific	Exact: Changeup, Right
Curveball	Left	Pooled-Supported	Scored	Group-specific	Exact: Curveball, Left
Curveball	Right	Rejected	Scored	Group-specific	Exact: Curveball, Right
Cutter	Left	Rejected	Scored	Group-specific	Exact: Cutter, Left
Cutter	Right	Rejected	Scored	Group-specific	Exact: Cutter, Right
Four-Seam	Left	Published	Scored	Group-specific	Exact: Four-Seam, Left
Four-Seam	Right	Published	Scored	Group-specific	Exact: Four-Seam, Right
Sinker	Left	Rejected	Scored	Group-specific	Exact: Sinker, Left
Sinker	Right	Rejected	Scored	Group-specific	Exact: Sinker, Right
Slider	Left	Pooled-Supported	Scored	Group-specific	Exact: Slider, Left
Slider	Right	Rejected	Scored	Group-specific	Exact: Slider, Right
Splitter	Left	Pooled-Supported	Scored	Group-specific	Exact: Splitter, Left
Splitter	Right	Rejected	Scored	Group-specific	Exact: Splitter, Right
Two-Seam	Left	Rejected	Scored	Group-specific	Exact: Two-Seam, Left
Two-Seam	Right	Pooled-Supported	Scored	Group-specific	Exact: Two-Seam, Right

Calibration, Drift, and Sparse Samples

Raw tree probabilities are not automatically well calibrated. The model therefore uses Platt scaling, which fits a logistic calibration model to the prediction logits. Calibration is cross-fitted by pitcher during 2025 validation. A pitch type and handedness combination receives its own calibrator when the sample is large enough; otherwise it uses a pooled league calibrator. My code saves calibration bins comparing mean predicted whiff probability with observed whiff rate so that an apparently strong ranking can be checked against probability quality.

The untouched 2026 data is also checked for distribution shift. Numeric features are reviewed with standardized mean differences, population stability index, and changes in missing data rates. Categorical features are checked for changes in their level shares. Stable, Monitor, and Review flags make it easier to separate a true model problem from a season in which the pitch mix or tracking environment changed.

Sparse groups are handled through transparent fallbacks instead of being silently combined or dropped. The preferred Stuff+ reference is the exact 2025 pitch type and pitcher handedness group. If that group does not contain enough pitches, the model can pool the pitch across hands or move to the global pool. The chosen source and reference pitch count are written to the output. This keeps coverage broad while making it clear when a score relies on more pooled information.

Converting Neutral Whiff Probability to Stuff+

The location neutral probability is converted to a familiar plus scale using 2025 reference data. Within each pitch type and pitcher hand reference, the model takes the logit of neutral whiff probability, centers it on the reference mean, and divides by the reference standard deviation. The standardized value is then placed on a scale where 100 is average and 10 points represent one standard deviation. Pitch level scores are capped between 40 and 160 to prevent a small number of extreme predictions from dominating a report.

$$\text{Stuff+} = 100 + 10 \times [(\text{logit of neutral whiff probability} - 2025 \text{ reference mean}) / 2025 \text{ reference SD}]$$

Pitch type summaries are pulled toward 100 with a 30 pitch reliability prior. A three pitch changeup should not be presented with the same certainty as a 150 pitch fastball. Overall pitcher Stuff+ is calculated once across all of the pitcher's eligible pitches and receives one pitcher level reliability adjustment. The individual report and the team ranking use that same pooled definition and the same date window, while rankings are sorted on the unrounded score. This keeps the headline number consistent across every output.

Current Reporting Example

The current 2026 Falmouth output shows how the model can be used in a baseball operations setting. Aidan Keenan ranked first among the qualifying Falmouth pitchers at 110.9 Stuff+ over 93 pitches. His individual report identifies the four seam fastball as the driver of the arsenal at 114 Stuff+, with the cutter and slider slightly above average. The changeup was below average in the report, but it was also represented by only three pitches and was therefore pulled heavily toward the league mean. That combination of grade and sample size is more useful than presenting the pitch score by itself.



Aidan Keenan Stuff+ Report

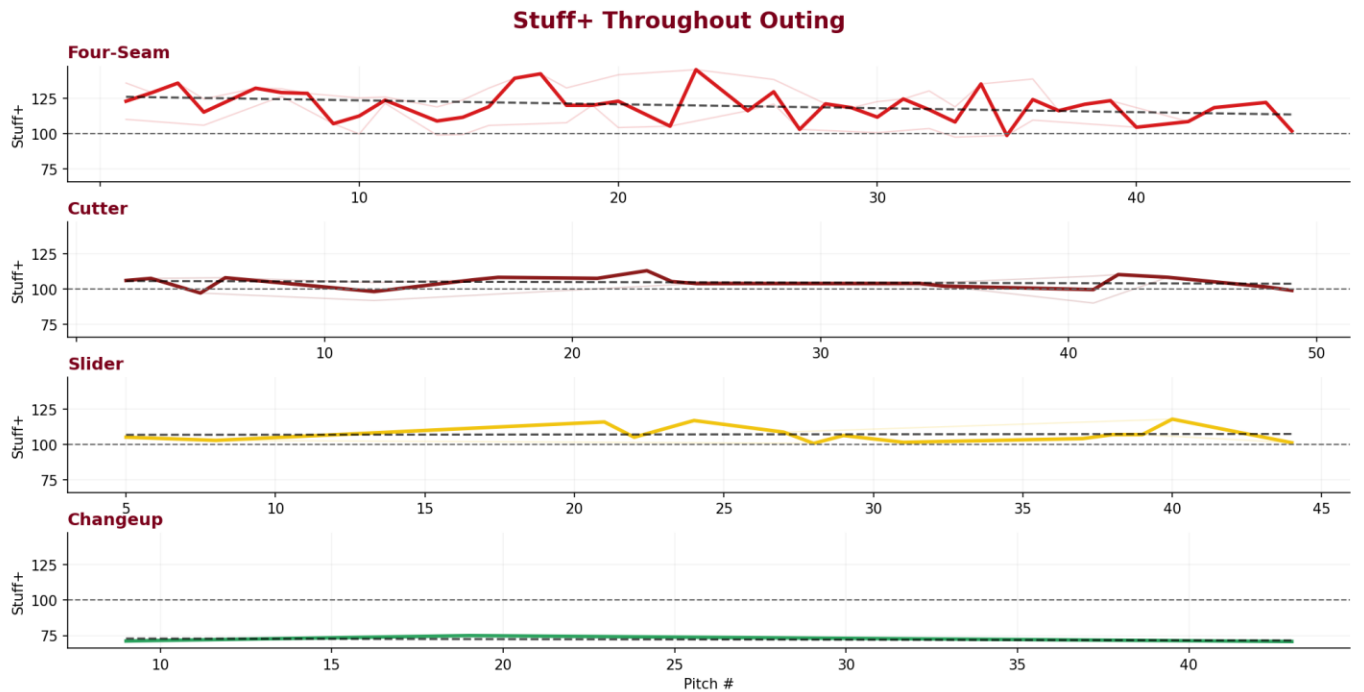
2026-06-10 to 2026-08-12



Overall Stuff+ 110

Pitch Arsenal Summary

Pitch	Total	Stuff+
Four-Seam	55	113
Cutter	21	102
Slider	14	102
Changeup	3	97



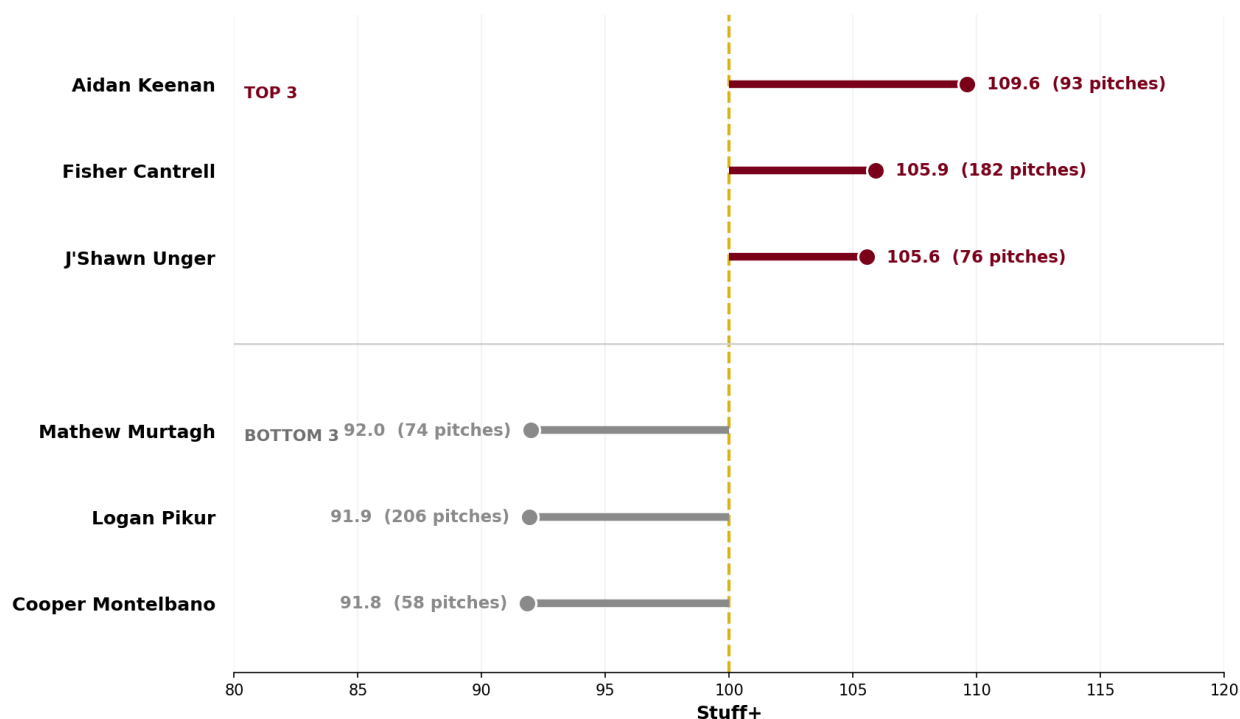
The same team graphic places Fisher Cantrell at 106.7 and Ben Reimers at 106.3 behind Keenan. The visual also shows the bottom three qualifying pitchers, using 100 as the league average reference line and a 25 pitch minimum for inclusion. This is not intended to be a public ranking of who is the best pitcher. It is a quick internal view of whose tracked

arsenal generated the strongest or weakest location neutral pitch quality scores during the selected window.



2026 Falmouth Commodores Pitcher Stuff+

Top 3 and Bottom 3 • 2026-06-10 to 2026-08-12 • Minimum 25 pitches • 100 = league average



Baseball Operations Value

A high score can lead an evaluator back to video to understand the pitch's shape, release, and hitter reactions. A low score can start a discussion about whether the pitch needs a different velocity band, movement target, grip, or usage plan. The line charts can also show whether a pitch held its shape through an outing or whether its quality changed as the pitcher fatigued. Because every pitch receives a score, the report can follow less common offerings without dropping them from the arsenal summary.

For roster construction or player acquisition purposes, Stuff+ can help create a consistent first pass across a large group of pitchers. It can flag an arm whose results were modest but whose raw pitch traits deserve another look, or it can show when strong results were supported more by location than by the underlying pitch. For player development, it gives coaches a repeatable baseline for testing adjustments. If a pitcher changes a grip or release cue, the staff can compare the new pitch shape and neutral score with his prior version rather than relying only on a handful of game outcomes.

Limitations and Next Steps

Stuff+ is intentionally narrower than total pitching value. It is trained on whiffs, so it does not directly reward called strikes, weak contact, ground ball quality, command, tunneling, sequencing, or the ability to execute in a specific count. A pitcher with average Stuff+ can still be highly effective through command and pitch design, while a pitcher with premium Stuff+ can underperform if he doesn't have the sharpest control or command. The model should be paired with Location+, outcome data, scouting looks, and video rather than treated as a standalone answer.

The model also depends on TrackMan measurement quality and pitch labels. Summer ball samples can be uneven, and the 2026 holdout comes from the same league environment as the training data. A future test against another league or a later CCBL season would provide a stronger measure of portability. The counterfactual neutral location grid is another modeling assumption. It reduces the influence of location, but it cannot perfectly recreate how every pitch would behave or be used in every part of the zone.

The next improvements could focus on monitoring rather than adding more questionable features. The current diagnostics already make it possible to review calibration, pitch group confidence, and seasonal drift. Future versions could add repeated out of season tests, confidence intervals around pitcher and pitch type summaries, and comparisons against a simpler velocity-and-movement baseline. Those steps would strengthen the evidence behind the model without moving it away from its main purpose, which is measuring the physical quality of a pitch as cleanly and transparently as possible.